Trainings

Theory of Operation

⚠ CAUTION ⚠

Do not replace main bearing shells without performing this inspection. Engine damage is possible if the main bearing shells are replaced without performing this inspection.

Fretting is micro-motion between the main bearing cap and block that results in material loss at the joint. This motion causes a loss of crankshaft to main bearing running clearance that can lead to a spun main bearing, a spun connecting rod bearing, and/or a broken crankshaft.

There is risk when overhauling or replacing the main bearings in a block that has incurred main cap fretting. Inserting new standard main bearing shells into a joint that has sustained sufficient loss of material can lead to zero crankshaft/bearing clearance. This can cause the crankshaft to lock in place.

Recent long-hour engineering tests have shown the addition of Loctite® 518, a block stiffener plate, and increased main bearing clamp loads reduces torsional motions in the main cap/block joint. Loctite® 518 adds an additional shear strength and seals the main bearing cap/block joint from oil and debris. The stiffener plate increases block stiffness by 15 percent and the new torque-plus-angle method increases main bearing clamp load by 1588 kg [3500 lb].

Note : The inspection must be performed with the original main cap and bearing shells. During the reduction of bearing running clearance, the original main bearing develops a wear pattern, which is an indicator of main cap fretting.

Plastigage™ provides a fast and accurate method to check bearing clearances. Plastigage™ is a special extruded plastic thread of a definite diameter with accurately controlled crush properties. Plastigage™ is packaged in calibrated envelopes, 12 envelopes per box. These envelopes **not only** protect the plastic threads but also serve directly as scales to measure the bearing clearance. Both sides of the envelope have a printed scale of graduations. One side is calibrated in inches, the other in millimeters. The numbers on the scale are the bearing clearances in thousandths of an inch or millimeter. When the width of a compressed section of Plastigage™ in a bearing or journal is compared with the appropriate-numbered graduation, the bearing clearance can be read directly from the scale.

Plastigage™ is available from a local parts store in four styles to cover different clearance ranges. Both Plastigage™ thread and its matching envelope have a distinctive color for each clearance range.

Acceptance limits were developed to make sure of proper oil film thickness between the main bearings and crankshaft. The minimum acceptable clearance is 0.05 mm [0.002 in] for the ISM series engines. The maximum acceptable clearance is 0.13 mm [0.005 in] for the ISM series engines. Blocks **not** condemned can be up-fitted with the block stiffener plate.

The block meets acceptance limits if the main bearing bore passes inspection and can be reassembled with new standard bearing shells if all of the following conditions exist:

- Main cap and block mating surfaces exhibit little or no fretting
- Lower main bearing shells (2 through 6) do **not** show copper exposure or uneven wear
- Plastigage™ (if done) is equal to or greater than 0.05 mm [0.002 in] and equal to or less than 0.13 mm [0.005 in].

The block main bearing bore does **not** pass inspection when the two following conditions exist:

- Main cap and block mating surfaces exhibit fretting
- Plastigage™ clearance on **any** main bearing (2 through 6) is less than 0.05 mm [0.002 in] or greater than 0.13 mm [0.005 in].

Repair options for blocks with one main journal that do **not** meet acceptance limits and have **not** spun a main bearing:

- Install an undersized main bearing kit to incrase bearing clearance.

Repair options for blocks with multiple main journals that do **not** meet acceptance limits or have spun a main bearing:

- Replace block with service block kit
- Replace block with ReCon® service block kit
- Replace block with ReCon® short block
- Replace engine with ReCon® engine.

Note : Cummins Inc. must approve all warrantable block repairs.

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